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(54) **MULTI-DIAPHRAGM VITRECTOMY PROBE**

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CPC **A61F 9/00763** (2013.01)

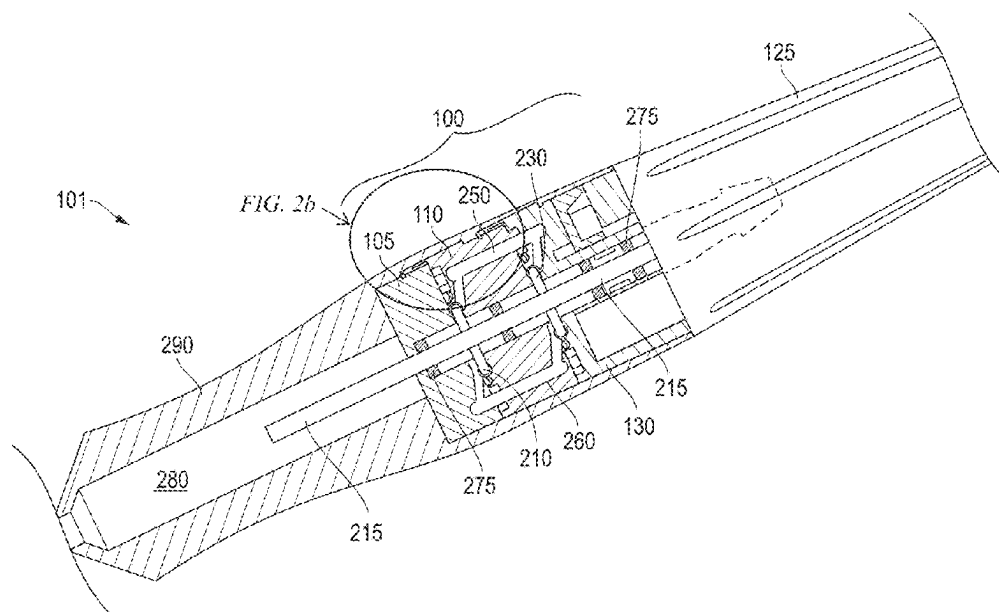
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See application file for complete search history.

(57) **ABSTRACT**

A diaphragm driven vitrectomy probe employing multiple diaphragms. The probe includes multiple diaphragms providing added areas to which hydraulic air may be delivered in reciprocating a cutter of the probe. Thus, the diaphragms and the probe housing may be reduced in size without sacrifice to force driving cutter reciprocation. Additionally, the amount of force attained and the rate of cutter reciprocation may be increased through the use of multiple diaphragms. Alternatively, the use of added diaphragms may allow for the rate of cutter reciprocation to be maintained even while the air pressure utilized may be reduced.

10 Claims, 6 Drawing Sheets



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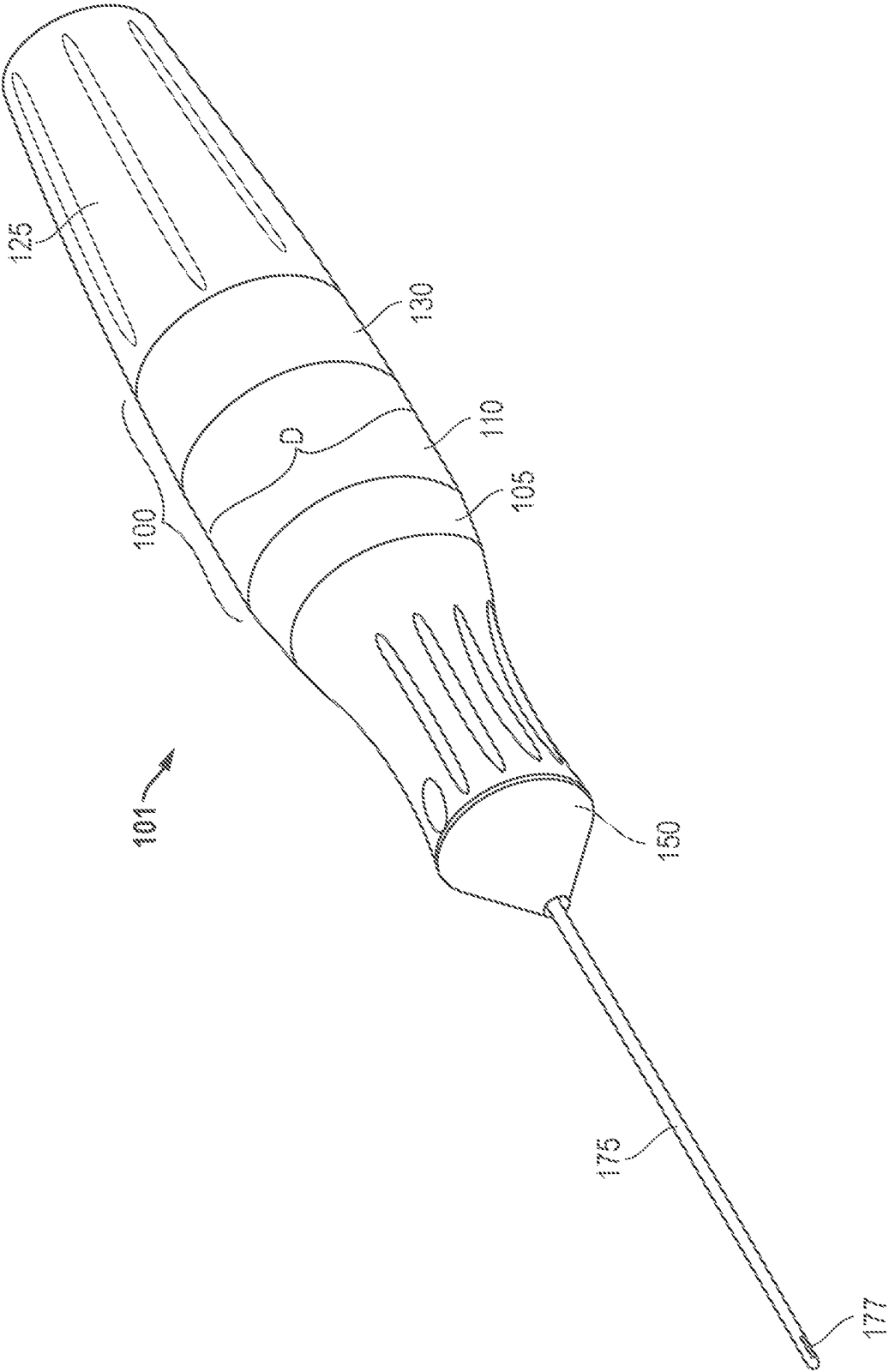


FIG. 1

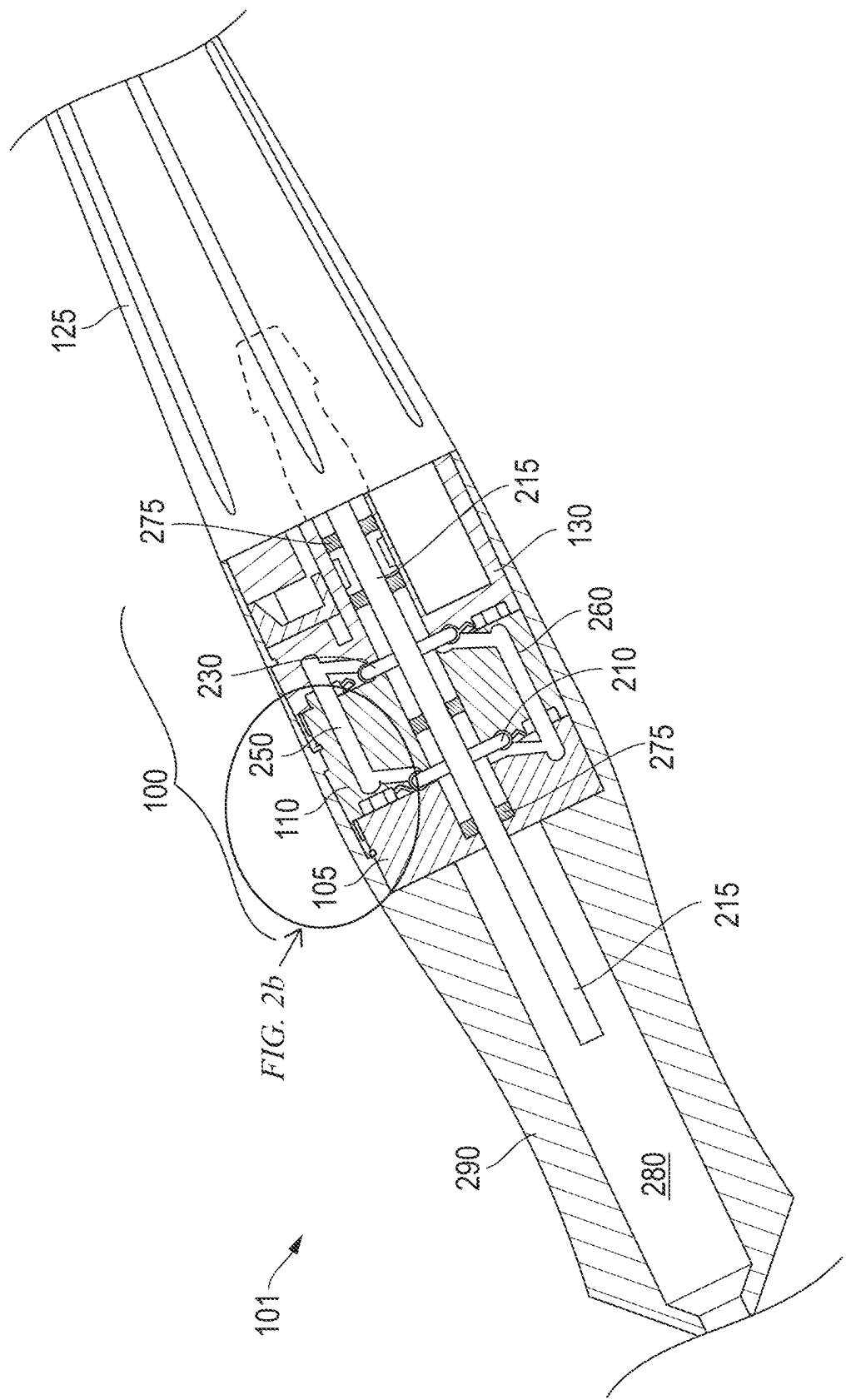
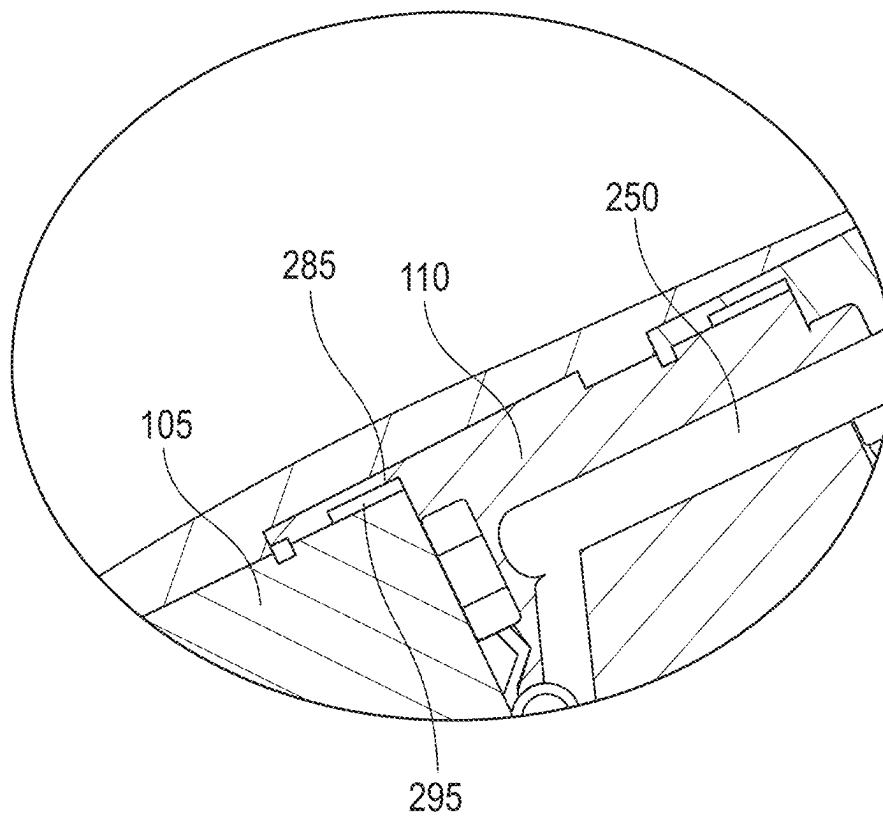


FIG. 2a

*FIG. 2b*

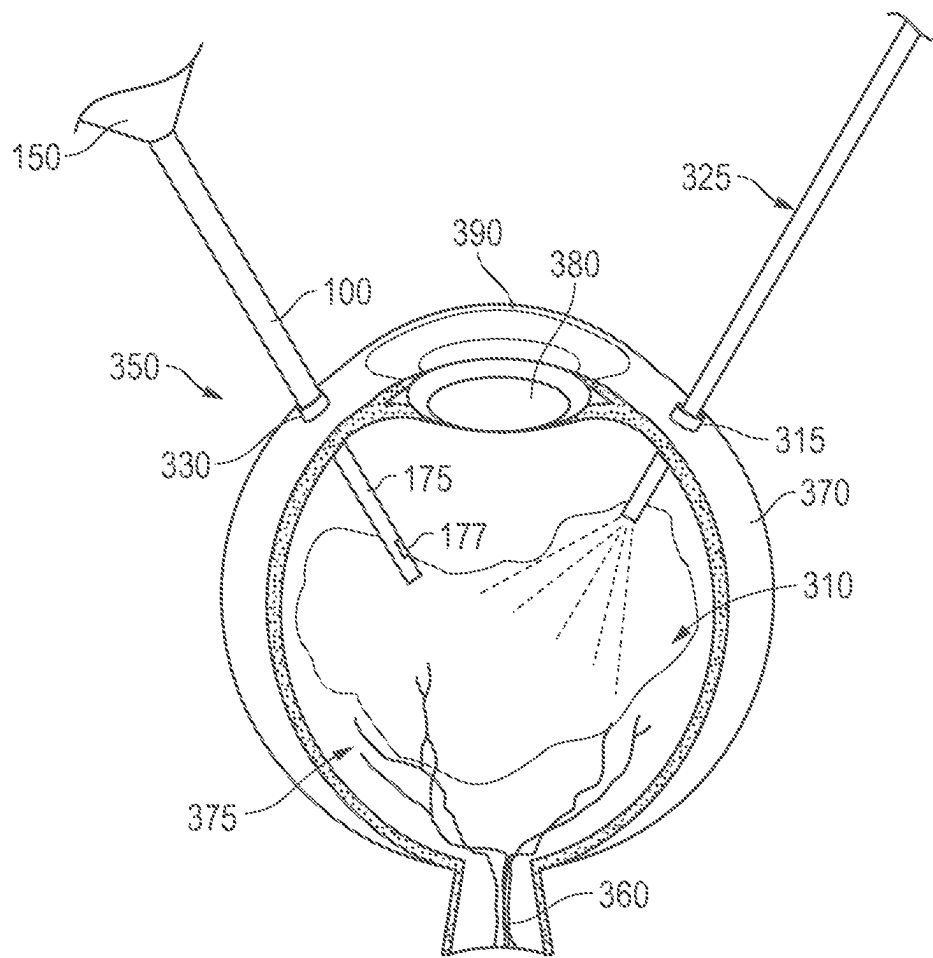


FIG. 3

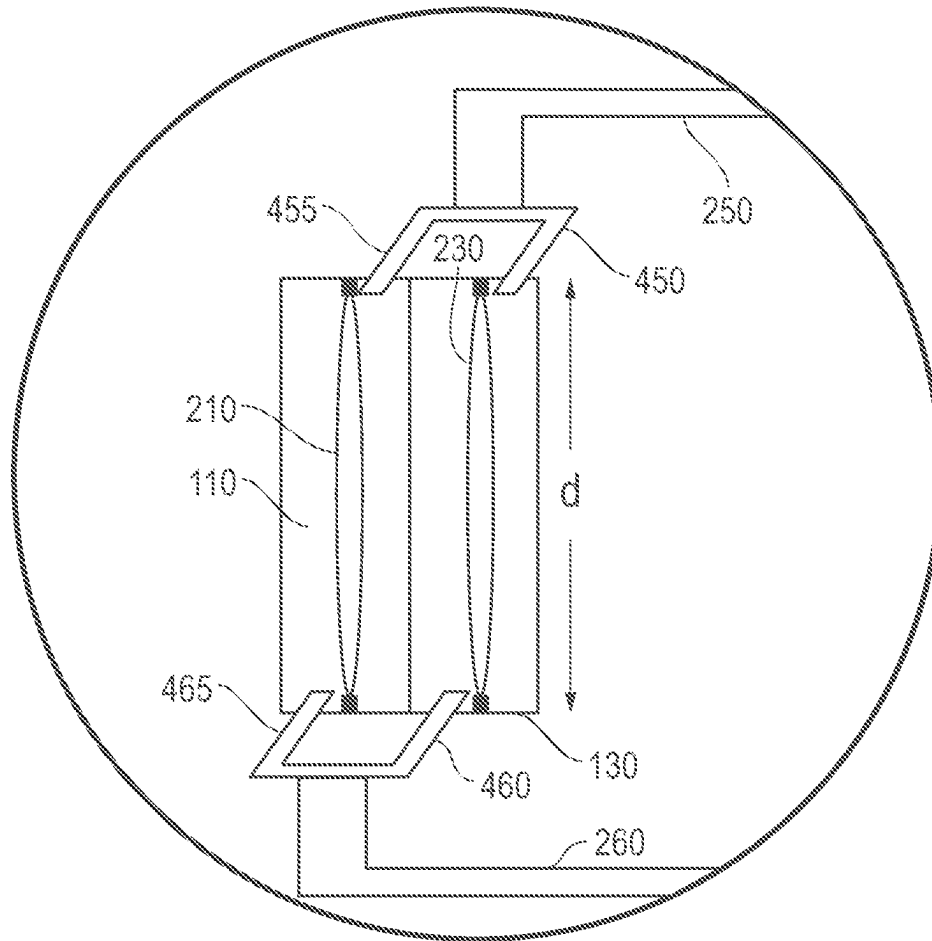
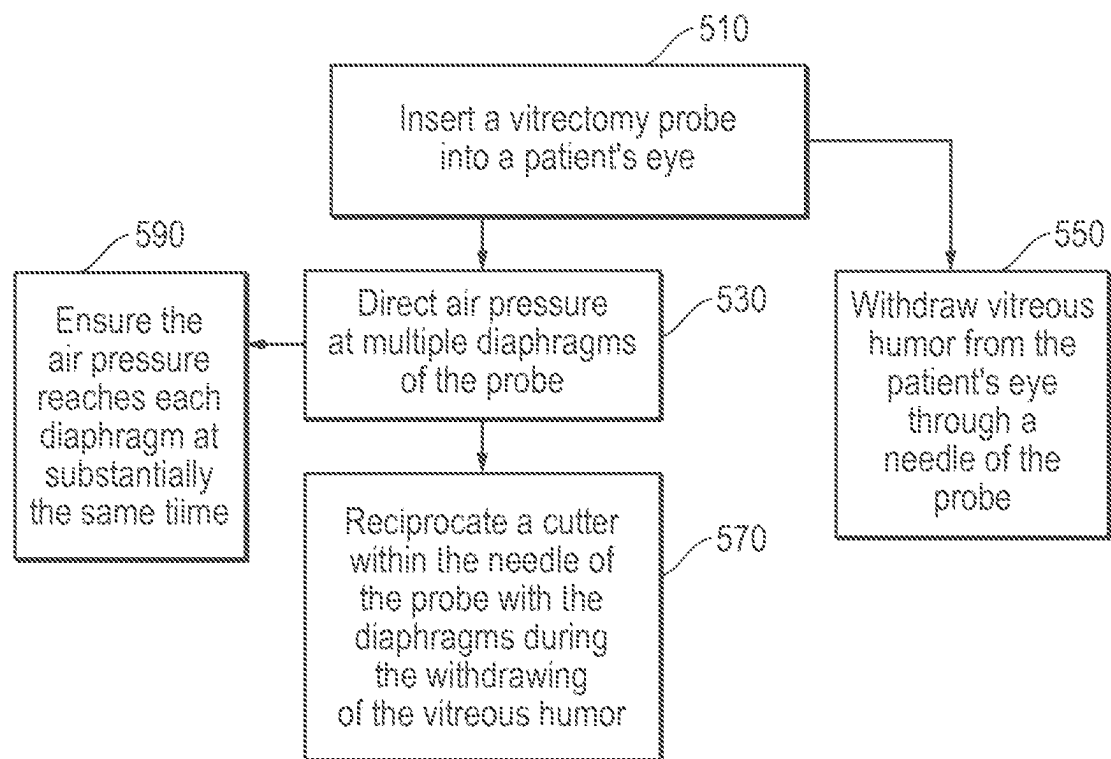


FIG. 4

*FIG. 5*

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MULTI-DIAPHRAGM VITRECTOMY PROBE**PRIORITY CLAIM**

This application claims the benefit of priority of U.S. Provisional Patent Application Ser. No. 63/088,529 titled “MULTI-DIAPHRAGM VITRECTOMY PROBE,” filed on Oct. 7, 2020, whose inventors are Nathaniel Reyes, Jesus R. Gonzales, Jr., and Mark Vojtasek, which is hereby incorporated by reference in its entirety as though fully and completely set forth herein.

BACKGROUND

Over the years, many dramatic advancements in the field of eye surgery have taken place. When the eye surgery includes accessing the retina, it is common that a vitrectomy will be included in at least part of the procedure. Vitrectomy is the removal of some or all of the vitreous humor from a patient’s eye. In some cases, where the surgery was limited to removal of clouded vitreous humor, the vitrectomy may constitute the majority of the procedure. However, a vitrectomy may accompany surgery to repair a retina, to address a macular pucker, or a host of other issues.

The vitreous humor itself is a clear gel that may be removed by an elongated probe when inserted through a pre-placed cannula at the eye. More specifically, the probe includes a central channel for removal of the vitreous humor. Further, the cannula provides a structurally supportive conduit strategically located at an offset location at the front of the eye, such as the pars plana. In this way, the probe may be guidingly inserted into the eye in a manner that avoids damage to the patient’s lens or cornea.

Unfortunately, removal of the vitreous humor requires greater care than simply applying a vacuum through the channel of the probe. This is because the vitreous humor includes a fibrous matrix of collagen fibrils. Therefore, merely applying a vacuum to the gel would place the surrounding eye structure in jeopardy. That is, the fibrous nature of the gel is such that a vacuum pull on the gel into the probe might translate into a pull on the retina, optic nerve or other delicate eye structures.

In order to address this issue, vitrectomy probes are configured to cut vitreous humor as it is drawn into the channel of the probe. In this way, a continuous fibrous pull on the gel-like substance does not translate into a pull on delicate eye structures. Instead, the vitreous humor is pulled into the channel of the probe in very small, chopped segments. This chipping or cutting of the vitreous humor occurs by the reciprocation of a cutter within the channel of the probe. More specifically, the cutter reciprocates back and forth at a port for intake of the vitreous humor in a manner that cuts the substance as it is being drawn into the channel. Perhaps 5,000 to 10,000 cuts per minute may take place in this manner in order to safeguard the eye from pulling by the vitreous humor as it is being removed. In fact, depending on the internal architecture of a reciprocating diaphragm, the cutter may achieve up to 15,000 cuts per minute (or higher). For example, this may be the case where a diaphragm having an effective diameter of a little over 0.41 inches is employed. Once more, this may be doubled to about 30,000 cuts per minute (or higher) where a two-way cutter is utilized, wherein each reciprocation results in two cuts, one in each direction of the reciprocation.

Of course, reciprocating a cutter by way of a reciprocating diaphragm that is over about 0.40 inches in effective diameter and accounting for the housing and other architecture

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built up around the diaphragm, the probe may have an outer diameter that is well over about 0.6 inches, often up to 0.7 inches. By way of comparison, consider a very large sharpie or marker. While the surgeon may prefer the option of tool with a diameter the size of a pencil for sake of added control during a vitrectomy procedure, such an option may simply not be practical due to the underlying size of the diaphragm. Ultimately, the size of the diaphragm presents a design limitation to the probe in terms of final diameter.

Of course, the size of the diaphragm may be reduced in order to reduce the ultimate probe diameter. Indeed, for tools where no diaphragm is required, such as a laser instrument, the diameter is often less than about 0.3 inches, closer to that of a pencil or other precision instrument. However, applying this type of thinking to a vitrectomy probe and minimizing the diaphragm size would result in a reduction in force that the diaphragm is able to impart on the cutter and thus a fairly dramatic reduction in cutter speed. In fact, reducing the size by about 30% would cut the force in half. The end result would be to dramatically reduce the performance of the probe function. Ultimately, with current probe technology, the surgeon is left with either an instrument less capable in terms of performance or a larger diameter instrument which may afford the surgeon less precision.

SUMMARY

A vitrectomy probe is disclosed. The probe includes a first diaphragm and a second diaphragm. Each diaphragm is driven in a first direction and an opposite second direction by hydraulic air that is reciprocatingly delivered to each. In this way, two separate diaphragms in a series are simultaneously employed to reciprocate the same cutting support of the probe during use in a surgical procedure.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of an embodiment of a multi-diaphragm vitrectomy probe.

FIG. 2a is cross-sectional view of the multi-diaphragm vitrectomy probe of FIG. 1 revealing unique air channels thereto.

FIG. 2b shows an alternate configuration for using snap fitting for connecting segments of the vitrectomy probe together.

FIG. 3 is a side cross-sectional overview of a patient’s eye during a vitrectomy procedure in which the multi-diaphragm vitrectomy probe of FIGS. 1 and 2 is utilized.

FIG. 4 is schematic view of an embodiment of an architectural layout for the air channels of FIG. 2a.

FIG. 5 is a flow-chart summarizing an embodiment of utilizing a multi-diaphragm vitrectomy probe during a vitrectomy procedure.

DETAILED DESCRIPTION

In the following description, numerous details are set forth to provide an understanding of the present disclosure. However, it will be understood by those skilled in the art that the embodiments described may be practiced without these particular details. Further, numerous variations or modifications may be employed which remain contemplated by the embodiments as specifically described.

Embodiments are described with reference to certain types of vitrectomy probe surgical procedures. In particular, a procedure in which vitreous humor is removed to address vitreous hemorrhage is illustrated. However, tools and tech-

niques detailed herein may be employed in a variety of other manners. For example, embodiments of a vitrectomy probe as detailed herein may be utilized to address retinal detachments, macular pucker, macular holes, vitreous floaters, diabetic retinopathy or a variety of other eye conditions. Regardless, so long as the vitrectomy probe incorporates multiple diaphragms in series, appreciable benefit may be realized.

Referring now to FIG. 1, a perspective view of an embodiment of a multi-diaphragm vitrectomy probe **101** is illustrated. The probe **101** includes a component housing **100** that is segmented. Specifically, distal **105**, middle **110** and proximal **130** housing segments are illustrated. These segments **105**, **110**, **130** may or may not be visible from the exterior. For example, with added reference to FIG. 2a, they may be located beneath a housing cover **290**. Regardless, the segments **105**, **110**, **130** are configured to accommodate diaphragms **210**, **230** therebetween. Thus, as suggested above, the probe **101** may be referred to as a multi-diaphragm vitrectomy probe **101**.

The housing **100** is coupled to a shell **125** which is provided as an ergonomic support for a surgeon employing the probe **101** during a procedure. In absence of the shell **125**, the handheld portion of the probe **101** effectively consists of no more than the housing **100** which may be under a few inches in total length. A surgeon may or may not choose to utilize the probe **101** with the shell **125** in place as illustrated. That is, as a matter of user preference, the surgeon may choose to remove the shell **125** for surgery. Thus, the probe **101** is configured such that the shell is removable in a user friendly manner that does not subject the probe **101** to potential damage with the surgeon crudely attempting to pry the shell **125** from the probe **101**. In this way, the vitrectomy procedure may be performed with the surgeon holding the end casing **150** solely between a thumb and forefinger without any other interfering support.

Returning to the component housing **100**, notice that it is of a given diameter (D). As discussed further below, the diameter (D) may be reduced to a degree depending on the number of segments **105**, **110**, **130** utilized which is in turn based on the number of diaphragms **210**, **230** utilized (see FIG. 2a). For example, in one embodiment, the diameter (D) may be about half an inch while still generating the same force that might result from a conventional component housing that is well in excess of 0.60 inches in diameter. Thus, the probe **101** itself may become thinner and potentially more maneuverable for the surgeon without sacrifice to cut rate or performance. Indeed, the flexibility in design may allow for even greater cut rate and performance such as where more than two diaphragms are utilized or the use of a housing cover **290** of any substantial thickness is avoided (see FIG. 2a). More specific exemplary embodiments and numbers are provided below. Regardless, so long as multiple diaphragms **210**, **230** are employed, the probe **101** may be made thinner without any sacrifice to performance, and indeed, performance may even be improved (again see FIG. 2a). Further, to maintain traditional dimensions where the overall probe length from end casing **150** to the opposite end of the shell **125** at about 3 inches or less, the shell **125** may simply be reduced in size as more and more segments **110**, **130** are added. In an embodiment where each segment **105**, **110**, **130** takes up between about 0.1 and 0.3 inches, perhaps up to five segments **105**, **110**, **130** (e.g. 4 diaphragms) may be utilized without adding to the overall length of the probe **101**. Of course, the option of allowing the overall length of the probe **101** to increase may also be viable.

Referring more directly now to FIG. 2a, a cross-sectional view of the multi-diaphragm vitrectomy probe **101** of FIG. 1 is illustrated. In this view, air channels **250**, **260** of fairly unique architecture are visible. Specifically, as discussed further below, the front channel **260** is configured to ensure that air directed at the front side of the diaphragms **210**, **230** reaches both diaphragms **210**, **230** at substantially the same time. Further, the back channel **250** is configured to ensure that air directed at the back side of the diaphragms **210**, **230** reaches both diaphragms **210**, **230** at substantially the same time.

Air reaching multiple diaphragms **210**, **230** simultaneously and in a reciprocating manner reciprocates an extension tube **215** which accommodates the vitreous humor cutter within a passage **280** as referenced above. The force that drives this reciprocation is a combined force obtained from each of the reciprocating diaphragms **210**, **230**. More specifically, the force generated is equal to the supplied air pressure multiplied by the area for each of the diaphragms **210**, **230**. So, for example, where 10 psi is applied to a conventional larger diaphragm with a diameter of about 0.41 inches, a force of about 1.32 lbs. might be obtained which might generally translate to between about 10,000-15,000 reciprocations per minute. In other words, $10 \times \pi (0.205)^2$ is 1.32 lbs. This may translate into 20,000-30,000 cuts per minute where the probe employs a double cutter (with cuts in both directions of the reciprocation). Regardless, in the embodiment shown, the diaphragms **210**, **230** may be smaller than a conventional diaphragm, perhaps about 0.29 inches in diameter. Ultimately, this may result in a thinner probe **101** as noted above. Nevertheless, because there are multiple diaphragms **210**, **230**, there need not be any sacrifice to the force attained. More specifically, $10 \times \pi (0.145)^2$ is 0.66 lbs. for each of two diaphragms **210**, **230**. Thus, 1.32 lbs. of force is still attained in total which should still translate into between about 10,000-15,000 reciprocations per minute.

Of course, the amount of force is not the only factor that determines the reciprocation rate. For example, the extension tube **215** interfaces a variety of seals **275** that are employed to ensure discrete pressure isolation during reciprocation as described. This may affect the rate depending on the degree of force at the interfaces between the seals **275** and the tube **215**. However, by way of contrast to a conventional probe **101** with a larger diaphragm, with all other factors such as seal interfacing being the same, the utilization of smaller diaphragms has not sacrificed attainable force nor reciprocation rate such as in the example noted above.

Continuing with reference to FIG. 2a, note the modular nature of the probe housing **100**. In some embodiments, the segments **105**, **110**, **130** may be secured through a friction fit and/or adhesive for a smooth interface between the segments (e.g., as seen in FIG. 2a). In an alternate embodiment, the segments **105**, **110**, **130** are of an architecture for snap-fitting or mechanically keying together (e.g., as seen in the alternate embodiment shown in FIG. 2b). In some embodiments, the snap-fitting and/or mechanical keying of the different segments may be used along with, for example, adhesive (or other connection mechanisms) for a stronger bond between the sections. For example, snaps **285** may extend from one section and snap into an adjacent section. The material for the snap **285** may be the same material as the section it extends from and may be resilient enough to hold downward pressure on a keying feature at the end of the snap **285** that is retained in a corresponding recess in the adjacent section to hold the sections together. In some embodiments, the snap **285** may act as a skirt to retain adhesive in a pocket **295**. The

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pocket **295** may provide containment for adhesive applied between the sections (that may have been displaced during assembly of the sections). The additional adhesive in the pocket **295** created by the snap **285** may also provide a stronger bond between the sections. This, along with the manner of assembling the housing cover **290** thereover, may result in some added bulk to the overall probe **101**. However, even with these features, the reduction in diameter size for the diaphragms **210**, **230** may still result in a thinner diameter probe **101** and housing **100**.

While the above embodiment is tailored to reducing housing diameter, multi-diaphragm architecture may be utilized for other enhancements as well. For example, given the cumulative effect on force that results, a multi-diaphragm configuration may be utilized with conventional diameter sizing that does not provide a thinner probe **101**. Instead, forces may be driven upward beyond conventionally attainable lbs. without the requirement of increasing pressure beyond industry standards. Alternatively, conventional sizing may be employed with a multi-diaphragm configuration and air pressure reduced while still attaining the same total force and presumed reciprocation rate.

Referring now to FIG. 3, a side cross-sectional overview of a patient's eye **350** is shown during a vitrectomy procedure. During this surgical procedure, the vitrectomy probe **101** of FIGS. 1 and 2a is utilized. Specifically, the needle **175** is inserted through a preplaced cannula **330** and directed toward a region **310** where vitreous humor is to be removed. Specifically, as described above, a suction applied to port **177** is used for the uptake of the vitreous humor or other substances. For example, in the procedure illustrated, a hemorrhage may be taking place in the region **310** such that blood is drawn into the port **177** along with the vitreous humor.

As also described above, a cutter is reciprocating within the needle **175** during this delicate procedure. With added reference to FIG. 2a, this means that multiple diaphragms **210**, **230** are utilized to simultaneously generate the driving force for the reciprocation. As a result, the diameter of the probe **101** may be thinner for enhanced control and maneuverability. By the same token, reciprocation may be increased for a more fluid-like uptake of the vitreous humor or pressure utilized in driving the reciprocation may even be reduced without any reduction in cut performance.

Continuing with reference to FIG. 3, the surgery illustrated includes the probe **101** and a light instrument **325** reaching into the eye **350** through cannulas **315**, **330** positioned in an offset manner at the sclera **370**. In this way, the more delicate cornea **390** and lens **380** may be avoided. By the same token, the optic nerve **360** and retina **375** are also quite delicate. Therefore, given that the needle **175** is capable of reaching these delicate features at the back of the eye **350**, the thinner probe **101** with enhanced control and maneuverability may be of particular benefit.

Referring now to FIG. 4, a schematic view of an embodiment of an architectural layout for the air channels **250**, **260** of FIG. 2a is illustrated. In this view, the reduced diameter (d) of the diaphragms **210**, **230** is apparent with the diaphragms **210**, **230** stacked next to each other, at a location that is distal the source of the air pressure. That is, each channel **250**, **260** is fed by an air pressure source to the proximal (or right) of the illustration. This means that if the back channel **250** were to linearly and directly interface each diaphragm **230**, **210** in a simple fashion without modification, the air within the channel **250** would reach the diaphragms **230**, **210** in sequence (first the back diaphragm **230** and then the front **210**). The same would be the case for the

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front channel **260** if not modified from a more linear, direct air path. The end result would likely be continuous misfiring and locking of the reciprocation, possibly rendering the probe ineffective.

As illustrated in FIG. 4, this potential for improper timing and misfiring may be avoided where the flow-paths of the channels **250**, **260** are modified from a simple linear architecture to one which ensures that air through either channel **250**, **260** reaches either diaphragm **230**, **210** at substantially the same time. In the embodiment illustrated, this is achieved with each channel **250**, **260** splitting into equidistant subchannels **450**, **455** and **460**, **465** before reaching the diaphragms **230**, **210**.

Notice that in the case of the front channel **260** for the embodiment shown, this means that the split into the equidistant front subchannels **460**, **465** occurs at a location beyond the back diaphragm **230**. In other words, the front channel **260** traverses the location of the back diaphragm **230** before splitting into the subchannels **460**, **465** at the middle housing segment **110**. This reflects the fact that the diaphragms **230**, **210** are stacked and proximal the air source which ultimately needs to reach beyond the locations of the diaphragms **230**, **210** in order to target the front sides thereof for reciprocating (to the right in the illustration of FIG. 4). The back channel **250** is tailored to reach the back sides of the diaphragms **230**, **210** to impart the opposite stroke of the reciprocation (to the left in the illustration of FIG. 4). Thus, the split to the back subchannels **450**, **455** may occur at the proximal housing segment **130** while still keeping the subchannels **450**, **455** minimal and equidistant in reaching the back sides of the diaphragms **230**, **210**. Therefore, proper timing and reciprocation may be better ensured even in a multi-diaphragm configuration.

Referring now to FIG. 5, a flow-chart summarizing an embodiment of utilizing a multi-diaphragm vitrectomy probe during a vitrectomy procedure is illustrated. As with any such procedure, the tool is inserted into the patient's eye as indicated at **510** for withdrawing vitreous humor from the patient's eye by way of a needle of the probe (see **550**). For embodiments of the present application, at this time, air pressure is also directed at multiple diaphragms of the probe as noted at **530**. Thus, as indicated at **570**, a cutter within the needle of the probe is reciprocated by the diaphragms as the vitreous humor is drawn into the needle. Once more, the probe is configured to ensure that air pressure reaches each diaphragm at substantially the same time (see **590**).

Embodiments described hereinabove include techniques and configurations that allow for the thinning of a vitrectomy probe. Once more, this may occur without sacrifice to performance or cut rate of the probe. In addition, or alternatively, these same techniques and configurations may be employed to increase force and cut rate or even to decrease pressure utilized during a vitrectomy procedure. Ultimately, the use of a multiple diaphragm configuration allows for flexibility in design while allowing avoidance of sacrifice to vitrectomy probe performance.

The preceding description has been presented with reference to presently described embodiments. However, other embodiments and/or features of the embodiments disclosed but not detailed hereinabove may be employed. Furthermore, persons skilled in the art and technology to which these embodiments pertain will appreciate that still other alterations and changes in the described structures and methods of operation may be practiced without meaningfully departing from the principle and scope of these embodiments. Additionally, the foregoing description should not be read as pertaining only to the precise structures

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described and shown in the accompanying drawings, but rather should be read as consistent with and as support for the following claims, which are to have their fullest and fairest scope.

The invention claimed is:

1. A vitrectomy probe, comprising:
 - a first diaphragm secured about a reciprocating component, the first diaphragm driven in a first direction and an opposite second direction by air reciprocatingly delivered thereto; and
 - a second diaphragm secured about the reciprocating component, the second diaphragm driven in the first direction and the opposite second direction by the air reciprocatingly delivered thereto;
 wherein the air is simultaneously delivered to the first diaphragm and the second diaphragm to drive the first diaphragm and the second diaphragm in the first direction at the same time during a first half of the reciprocation, and subsequently, the air is simultaneously delivered to the first diaphragm and the second diaphragm to drive the first diaphragm and the second diaphragm in the second direction at the same time during a second half of the reciprocation.
2. The vitrectomy probe of claim 1, wherein the first diaphragm and the second diaphragm are positioned with a housing having a first diameter of under about 0.60 inches and the first diaphragm and the second diaphragm are of a second diameter of under about 0.30 inches.
3. The vitrectomy probe of claim 2, further comprising:
 - an end casing coupled to a first end of the housing for gripping by a surgeon during a vitrectomy procedure; and
 - a removable shell coupled to a second end of the housing opposite the first end to serve as an ergonomic support during the procedure.
4. The vitrectomy probe of claim 3, wherein the reciprocating component is an extension tube, the probe further comprising:
 - a needle emerging from the end casing to reach into an eye of a patient during the vitrectomy procedure; and
 - a cutter coupled to the extension tube for cutting vitreous humor taken into the needle during the vitrectomy procedure.
5. The vitrectomy probe of claim 1, further comprising:
 - a front air channel to supply the air to a front of each of the first diaphragm and the second diaphragm to support the driving of the first diaphragm and the second diaphragm in the first direction, the first direction being a distal direction; and
 - a back air channel to supply the air to a back of each of the first diaphragm and the second diaphragm to support the driving of the first diaphragm and the second diaphragm in a second direction, the second direction being a proximal direction.
6. The vitrectomy probe of claim 5, wherein the front air channel is split into equidistant subchannels to ensure that the supply of the air to the front of the first diaphragm and the second diaphragm reaches each of the first diaphragm

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and the second diaphragm simultaneously and the back air channel is split into equidistant subchannels to ensure that the supply of the air to the back of the first diaphragm and the second diaphragm reaches each of the first diaphragm and the second diaphragm simultaneously.

7. A segmented housing assembly for incorporation into a vitrectomy probe, the housing assembly comprising:

a distal segment and a middle segment to accommodate a first diaphragm secured about a reciprocating component therebetween, the first diaphragm driven in a first direction and an opposite second direction by air reciprocatingly delivered thereto; and

a proximal segment adjacent the middle segment to accommodate a second diaphragm secured about the reciprocating component therebetween, the second diaphragm driven in the first direction and the opposite second direction by the air reciprocatingly delivered thereto;

wherein the air is simultaneously delivered to the first diaphragm and the second diaphragm to drive the first diaphragm and the second diaphragm in the first direction at the same time during a first half of the reciprocation, and subsequently, the air is simultaneously delivered to the first diaphragm and the second diaphragm to drive the first diaphragm and the second diaphragm in the second direction at the same time during a second half of the reciprocation.

8. The segmented housing assembly of claim 7, wherein the segments are attachable to one another by one of a snap fit and mechanical keying, wherein the at least one of the snap fit and mechanical keying further define a pocket to contain adhesive between the segments and the at least one of the snap fit and mechanical keying.

9. The segmented housing assembly of claim 7, further comprising:

a front air channel to supply the air to a front of each of the first diaphragm and the second diaphragm to support the driving of the first diaphragm and the second diaphragm in the first direction, the first direction being a distal direction, the front air channel comprising equidistant subchannels to facilitate the air reaching the front of each of the first diaphragm and the second diaphragm simultaneously; and

a back air channel to supply the air to a back of each of the first diaphragm and the second diaphragm to support the driving of the of the first diaphragm and the second diaphragm in a second direction, the second direction being a proximal direction, the back air channel comprising equidistant subchannels to facilitate the air reaching the back of each of the first diaphragm and the second diaphragm simultaneously.

10. The segmented housing assembly of claim 9, wherein the equidistant subchannels of the front air channel emerge from a split at about a location of the middle segment and the equidistant subchannels of the back air channel emerge from a split at about a location of the proximal segment.

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